

THE SCIENCE OF SAFE CLINICAL AI: CONJOINING NEURAL INFERENCE AND SYMBOLIC CONTROL

DATASCI290: NEUROSymbolic AI, SPRING 2026

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RELATED

■ Core medical AI models and diagnostic agents

- Chen, Z., Hernández Cano, A., Romanou, A., et al. (2023). MEDITRON-70B: Scaling medical pretraining for large language models. arXiv. <https://doi.org/10.48550/arXiv.2311.16079>
- McDuff, D., Schaeckermann, M., Tu, T., et al. (2025). Towards accurate differential diagnosis with large language models. *Nature*, 642(8067), 451–457. <https://doi.org/10.1038/s41586-025-08869-4>
- Sellergren, A., Kazemzadeh, S., Jaroensri, T., et al. (2025). *MedGemma technical report*. arXiv. <https://doi.org/10.48550/arXiv.2507.05201>
- Singhal, K., Tu, T., Gottweis, J., et al. (2025). Toward expert-level medical question answering with large language models. *Nature Medicine*, 31, 154–166. <https://doi.org/10.1038/s41591-024-03423-7>
- Singhal, K., Azizi, S., Tu, T., et al. (2023). *Large language models encode clinical knowledge*. *Nature*, 620, 172–180. <https://doi.org/10.1038/s41586-023-06291-2>
- Tu, T., Schaeckermann, M., Palepu, A., et al. (2025). *Towards conversational diagnostic artificial intelligence*. *Nature*, 642(8067), 442–450. <https://doi.org/10.1038/s41586-025-08866-7>

■ Reasoning, verification, and medical LLM evaluation

- Dhuliawala, S., Komeili, M., Xu, J., Raheja, V., Li, M., & Boureau, Y.-L. (2023). *Chain-of-verification reduces hallucination in large language models*. arXiv. <https://doi.org/10.48550/arXiv.2309.11495>
- Sim, S. Z. Y., & Chen, T. (2025). Unveiling the reasoning behaviour of medical large language models. *eLife*, 14, e106187. <https://doi.org/10.7554/eLife.106187>
- Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. arXiv. <https://doi.org/10.48550/arXiv.2201.11903>
- Yun, J., Sohn, J., Park, J., et al. (2025). Med-PRM: Medical reasoning models with stepwise, guideline-verified process rewards. arXiv / EMNLP 2025. <https://doi.org/10.48550/arXiv.2506.11474>

■ Clinical decision support and regulatory context

- U.S. Food and Drug Administration. (2026). Clinical decision support software: Guidance for industry and Food and Drug Administration staff. U.S. Food and Drug Administration.
- U.S. Food and Drug Administration. (2025). Considerations for the use of artificial intelligence to support regulatory decision-making for drug and biological products: Draft guidance for industry and other interested parties. U.S. Food and Drug Administration.



WHAT SHOULD REGULATORY
REQUIREMENTS EVEN BE ?



MEDICAL AI DECISION SUPPORT SYSTEMS

- U.S. Food and Drug Administration. (2026). *Clinical decision support software: Guidance for industry and Food and Drug Administration staff*. <https://www.fda.gov/media/109618/download>
- **Human decision** authority: Clinical AI systems must support clinician judgment rather than replace it. The clinician must remain responsible for the final diagnostic or treatment decision.
- **Explainable** recommendation basis: The system must provide sufficient information about how recommendations are generated so that clinicians can independently evaluate the reasoning.
- Clear **intended clinical use**: The software must specify the intended users (e.g., physicians, nurses), clinical setting, patient population, and decision context in which the system is appropriate.
- **Transparent** input requirements: The system must clearly define the patient information and clinical data required to produce recommendations and how those inputs influence outputs.
- **Evidence-backed** development: Algorithms must be supported by clinical evidence, data sources, or validated medical knowledge used during development and training.
- **Regulatory** classification risk: Software that directly drives clinical decisions or replaces clinician judgment may be regulated as a medical device.

CLINICAL REASONING IS HARD !

- **Rapid progress in medical AI**

- Large language models trained on massive corpora
- Strong performance on clinical knowledge and medical QA
- Passing-level scores on licensing-style exams

- **However**

- Clinical **encounters** are fundamentally different
 - Conversational, incomplete, ambiguous patient descriptions
- Triage is a **dynamic reasoning** problem
 - Multiple competing hypotheses must be tracked
 - Evidence accumulates sequentially through questioning
 - Dangerous conditions must be ruled out early
- **Safety** dominates the problem
 - Early diagnostic errors are a major source of preventable harm



WHAT TYPES OF REASONING WOULD
BE REQUIRED IN CLINICAL AI
SYSTEMS ?



TOWARD STRUCTURED REASONING FOR MEDICAL AI

- Clinical reasoning involves **multiple reasoning modes**
 - Abductive reasoning
 - Generate plausible explanations for observed symptoms
 - Maintain a differential diagnosis
 - Deductive reasoning
 - Apply clinical rules, contraindications, escalation criteria
 - Detect red-flag conditions
 - Inductive reasoning
 - Use patterns learned from medical literature and prior cases
 - Sequential reasoning
 - Decide what information to gather next
 - Ask questions that reduce diagnostic uncertainty
- Limits of standalone LLM reasoning
 - Not verified
 - Can hallucinate
- Structured reasoning architecture
 - Neural models interpret narratives and generate hypotheses
 - Symbolic reasoning enforces safety constraints and rules

neural proposes, symbolic disposes

DIFFERENTIAL DIAGNOSIS

- What is **differential diagnosis**?
 - the clinical process of identifying and comparing multiple possible causes of a patient's symptoms.
 - clinicians generate a set of candidate conditions (the differential) and narrow it as evidence accumulates.
 - the goal is not to immediately find the “right answer,” but to rule out dangerous possibilities first.
- Clinical encounters are:
 - interactive
 - incomplete
 - language is “messy” - colloquial , vague, emotionally charged ...

DIFFERENTIAL DIAGNOSIS

- The **task is hypothesis management** (not answer generation !)
 - Maintain multiple competing diagnostic hypotheses.
 - Update the differential as new evidence appears or contradictions arise.
- Multiple reasoning modes operate simultaneously
 - Abductive reasoning: generate plausible explanations for observed symptoms.
 - Deductive reasoning: apply rules, contraindications, and escalation criteria.
 - Inductive reasoning: use patterns learned from prior cases and medical knowledge.
 - Sequential reasoning: decide what question or test should come next.
- Safety is ... “**asymmetric**”
 - Missing a dangerous condition is far costlier than being cautious.
 - Systems must emphasize red-flag detection, escalation, and abstention.
- Implication for AI system design
 - Differential diagnosis is far harder than medical question answering !
 - Strong performance on static QA or exam benchmarks does not demonstrate readiness for interactive clinical reasoning !

CURRENT AI MODELS (MEDICAL)

- Current landscape of medical AI systems
 - specialized medical LLMs: med-palm 2, meditron, medgemma, clinical-tuned llama variants
 - conversational diagnostic agents: amie and similar dialogue-optimized systems
 - general frontier LLM deployments in healthcare: GPT, Gemini, ...
- Recent progress: medical foundation models encode substantial clinical knowledge
 - med-palm 2 demonstrated strong medical QA and consultation responses
 - meditron and medgemma focus on open medical foundation models
 - amie pushes toward conversational diagnostic dialogue rather than static QA
- Benchmark strength $\neq$ Controlled clinical reasoning !
 - Interactive diagnosis is fundamentally different from QA
 - patient information appears gradually
 - evidence must be updated continuously
 - hypotheses must be revised over time
 - Static benchmarks capture only a small fraction of this process

CURRENT AI MODEL LIMITATIONS

- Pure neural reasoning remains vulnerable
 - confident but unsupported conclusions remain common
- Proposal and authorization are often blurred
 - models generate diagnoses and recommendations simultaneously
 - few systems enforce a computational boundary between suggestion and decision authority
- **Prompt-centric** system design
 - not **state-centric**
 - reasoning becomes unstable across long interactions
 - formal safety checks become difficult without structured state tracking
- Underuse of reasoning verification
 - recent research introduces process supervision and reasoning trace evaluation
 - intermediate reasoning steps can be checked explicitly
 - many diagnostic errors occur during intermediate inference
 - verification mechanisms remain underutilized in medical AI systems
- Implication
 - frontier medical models are powerful knowledge engines
 - but purely neural architectures remain insufficient for safe differential diagnosis support



STRENGTHS ?



LETS NOT IGNORE THE STRENGTHS

- Broad knowledge integration
 - trained on very large and diverse corpora
 - can synthesize information across many domains
 - strong recall of factual and conceptual knowledge
- Reasoning scaffolding
 - able to generate structured reasoning traces
 - can decompose problems into intermediate steps
 - useful for exploring candidate explanations or hypotheses
- Flexible problem solving
 - adapt to new tasks through prompting rather than retraining
 - perform multiple cognitive roles: analyst, summarizer, planner, explainer
- Dialogue and interaction
 - maintain multi-turn conversational context
 - ask follow-up questions and refine responses over time
 - support interactive reasoning workflows
 - overall natural language understanding

WHY NEUROSymbOLIC AI IS A PROMISING DIRECTION

- A better match to the structure of clinical reasoning
- A clinically useful diagnostic assistant must be able to:
 - Interpret messy patient language
 - Generate and revise diagnostic hypotheses
 - Enforce clinical constraints and safety rules
 - Determine when evidence is sufficient
- No single computational paradigm is naturally optimal for all of these tasks !

WHY NEUROSymbolic AI IS A PROMISING DIRECTION

- Matches the **heterogeneous reasoning demands** of medicine
 - Clinical diagnosis involves multiple reasoning modes:
 - abductive reasoning: generate candidate explanations
 - deductive reasoning: apply clinical rules and constraints
 - inductive reasoning: use statistical patterns from prior knowledge
 - sequential reasoning: reduce uncertainty through targeted questioning
 - A hybrid system can assign different roles to different mechanisms rather than forcing one model to do everything.
- Enables **verification of reasoning**
 - Structured representations allow reasoning to be inspected and checked.
 - Examples:
 - structured patient state
 - explicit evidence links
 - typed reasoning steps
 - Parts of the inference process can then be checked against:
 - knowledge sources
 - clinical guidelines
 - formal rules
 - This enables verifiable reasoning

WHY NEUROSymbOLIC AI IS A PROMISING DIRECTION

- Supports **asymmetric safety** requirements
 - Differential diagnosis is not a symmetric prediction task (certain errors are far more costly)
 - Symbolic constraints and action gating provide mechanisms for enforcing these safety requirements.
- **Scales better** than full symbolic medicine
 - The goal is not to encode all medical knowledge as rules.
 - Instead, formalize only what requires deterministic control:
 - red flags
 - contraindications
 - escalation rules
 - pathway gates
 - consistency checks
 - Everything else can remain neural:
 - narrative interpretation
 - hypothesis generation
 - exploratory reasoning

DESIGN

- **Purpose of the system**
 - Support early clinical reasoning when patients present with undifferentiated symptoms and incomplete information
 - Assist the reasoning process rather than automate diagnosis
 - Organize evidence, generate hypotheses, and identify when escalation or additional information is required
 - Maintain explicit mechanisms for safety, verification, and interpretability
- **Structured management of the differential diagnosis**
 - Maintain an explicit set of candidate diagnoses rather than a single prediction
 - Update hypotheses as new observations appear
 - Strengthen, weaken, or eliminate possibilities based on evidence
 - Use the differential to guide what evidence would most effectively distinguish between competing conditions
- **Identification of missing but clinically important information**
 - Recognize information gaps in the patient narrative
 - Determine which follow-up questions would most reduce diagnostic uncertainty
 - Prioritize questions that help rule out dangerous conditions
- **Early prioritization of serious conditions**
 - Clinical reasoning is asymmetric: missing a serious condition is far more costly than over-considering benign ones
 - Detect red-flag symptom patterns early
 - Trigger escalation or urgent evaluation when risk thresholds are crossed
- **Evidence-grounded reasoning outputs**
 - Link diagnostic hypotheses to the observations that support them
 - Maintain transparent reasoning that clinicians can inspect
 - Allow users to evaluate whether the reasoning process is appropriate

TECHNICAL GOALS

- **Reason over conversational clinical narratives**
 - Interpret patient descriptions expressed in everyday language
 - Extract clinically meaningful observations from dialogue
 - Maintain a structured patient state that evolves during the interaction
 - Capture symptoms, timing, negations, and contextual factors relevant to diagnosis
- **Verifiable intermediate reasoning**
 - Represent intermediate reasoning steps explicitly rather than implicitly
 - Evaluate whether proposed conclusions are supported by observed evidence
 - Check reasoning against extracted patient observations, medical knowledge sources, and known clinical constraints
- **Modular neurosymbolic system architecture**
 - Avoid a single monolithic model for clinical reasoning
 - Separate major functional components: conversational interpretation, patient state representation, hypothesis generation, knowledge retrieval, reasoning verification, constraint enforcement
 - Modular design enables improvements to individual components without redesigning the entire system
- **Separation of hypothesis generation and decision authorization**
 - Neural models generate candidate diagnostic hypotheses from narrative input
 - Verification and constraint mechanisms evaluate whether hypotheses are sufficiently supported
 - Only verified conclusions influence system recommendations

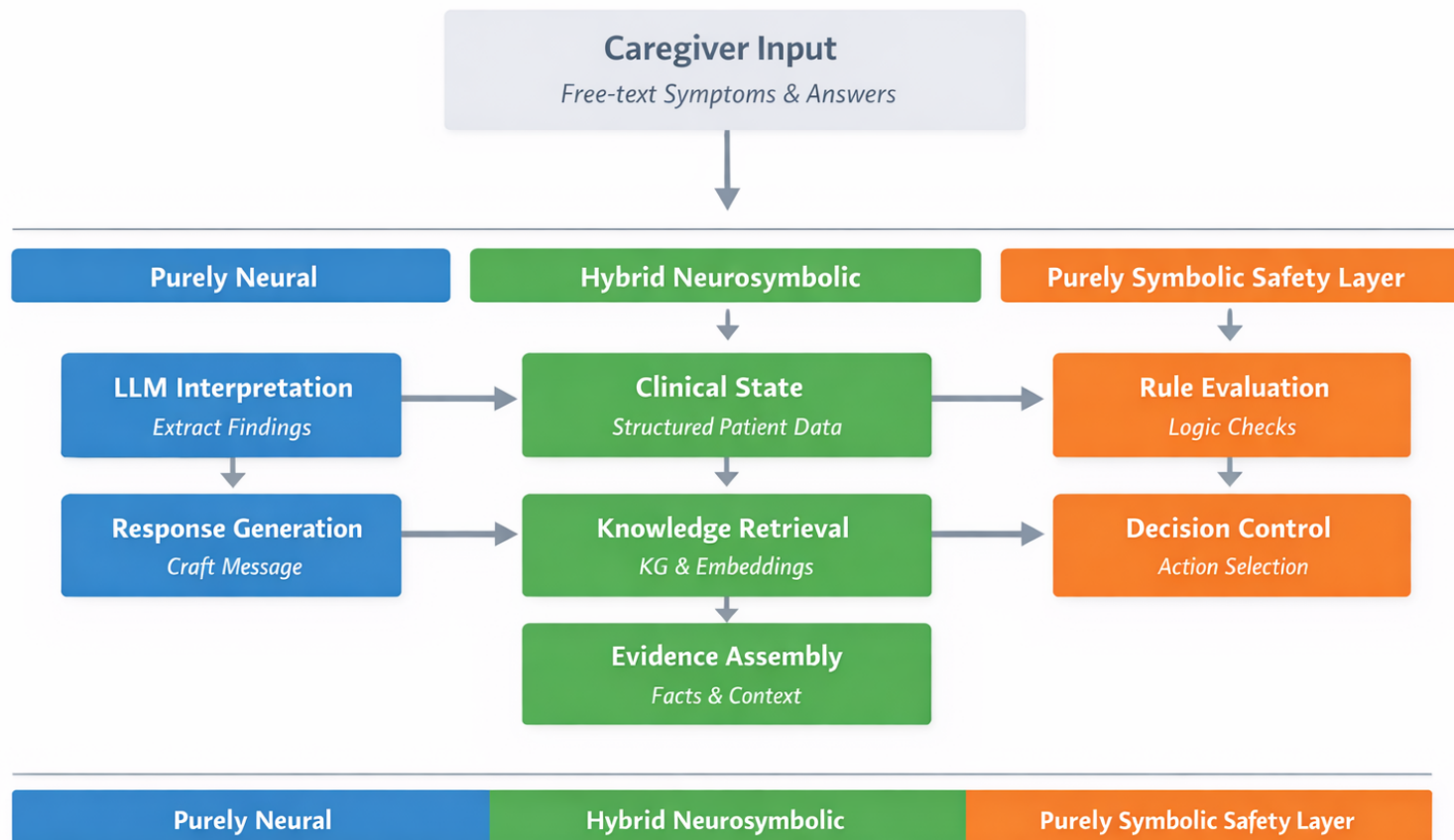
NEUROSymbolic AI System Design Principles

- **State-centric reasoning**
 - Maintain a persistent structured patient state that evolves during dialogue
 - Capture symptoms, timing, negations, context, and evolving diagnostic hypotheses
 - Enable reasoning over structured clinical state rather than raw conversation
- **Hypothesis-set reasoning**
 - Maintain and update a set of competing diagnostic hypotheses
 - Track evidence supporting or weakening each hypothesis
 - Preserve potentially serious conditions until sufficient evidence rules them out
- **Typed interfaces between reasoning components**
 - Neural models interpret language, structured state captures observations, knowledge graphs represent medical relationships, symbolic rules encode constraints
 - Typed interfaces ensure reliable information exchange and enable systematic reasoning checks
- **Selective formalization of clinical knowledge**
 - Encode only safety-critical elements deterministically: red flags, contraindications, escalation rules, pathway gates
 - Leave narrative interpretation and exploratory reasoning to neural models
- **Verification-first inference**
 - Evaluate candidate hypotheses against patient state, knowledge sources, and clinical constraints before influencing outputs
- **Modular model portfolio**
 - Separate models for conversational interpretation, reasoning generation, knowledge retrieval, and verification

TRUSTWORTHINESS, SAFETY, AND REGULATORY CREDIBILITY

- **Separation of hypothesis generation and decision authorization**
 - Neural models generate candidate interpretations and diagnostic hypotheses
 - Verification and constraint mechanisms determine which conclusions influence recommendations
 - Prevents fluent but unsupported reasoning from directly affecting clinical guidance
- **Explicit safety constraints**
 - Deterministic enforcement of safety-critical rules
 - Includes red-flag detection, escalation triggers, and contraindications
 - Implemented through explicit logic so they cannot be overridden by probabilistic outputs
- **Evidence-grounded reasoning**
 - Diagnostic suggestions linked to identifiable observations from patient interaction
 - May also reference external medical knowledge sources
 - Enables clinicians and evaluators to trace conclusions back to supporting evidence

ARCHITECTURE



REASONING & VERIFICATION FRAMEWORK

- Diagnosis as controlled reasoning
 - Differential diagnosis should not be treated as unconstrained answer generation
 - Hypotheses must be proposed, tested against evidence, checked against knowledge, and only then allowed to influence outputs
 - Emphasis shifts from final answer quality to process quality and reasoning control
- Coupling reasoning with verification
 - Neural models generate and update candidate hypotheses
 - Intermediate reasoning artifacts represent the evolving diagnostic process
 - Verification mechanisms evaluate these artifacts before recommendations are produced
 - Principle: the system should reason flexibly but act conservatively
- Intermediate reasoning traces
 - Maintain explicit structured artifacts of the reasoning process
 - Examples: hypothesis set, supporting findings, contradicting evidence, missing discriminators, urgency assessment, next best question
 - These artifacts are typed and inspectable rather than free-form explanations
- Why intermediate reasoning matters
 - Many diagnostic errors occur during reasoning steps, not only in the final answer
 - Systems may anchor too early, ignore contradicting evidence, or ask low-value questions
 - Explicit reasoning traces allow reasoning quality to be inspected and evaluated

REASONING & VERIFICATION FRAMEWORK

- Selective opportunities for formal verification
 - Some reasoning components can be checked deterministically
 - Examples: patient state consistency, pathway conditions, safety constraints, action permissions
- Multi-stage verification strategy
 - Grounding verification: ensure findings and hypotheses are supported by patient observations or evidence
 - Consistency verification: detect contradictions, temporal conflicts, or unexplained shifts in reasoning
 - Knowledge and pathway validation: check candidate inferences against complaint pathways, guidelines, and clinical knowledge
 - Safety rule enforcement: apply red-flag detection, escalation rules, contraindications, and minimum evidence requirements
- Key insight
 - reasoning failures arise at different levels: evidential, logical, pathway-related, and safety-critical
 - trustworthy systems therefore require **layered verification** rather than a single check

CLINICAL KNOWLEDGE INTEGRATION

- Purpose of the clinical knowledge layer
 - normalize messy patient language into stable medical concepts
 - provide structured clinical context for reasoning and verification
 - enable safety enforcement, evidence grounding, and pathway reasoning
- SNOMED CT as the canonical terminology backbone
 - map patient language to canonical clinical concepts
 - “tight chest”, “pressure here” → chest pain
 - “feeling faint” → presyncope
 - normalize synonyms, abbreviations, and lay expressions
 - support reasoning over concept hierarchies
 - respiratory symptoms, neurologic findings, infectious conditions
 - provide stable identifiers used across patient state, KG facts, retrieval, and rules
- Utility
 - enables reliable evidence grounding
 - supports rule execution over concept classes
 - prevents reasoning fragmentation caused by inconsistent terminology

CLINICAL KNOWLEDGE INTEGRATION

- Complaint pathways
 - Diagnosis usually starts with a presenting complaint (e.g., chest pain, fever)
 - Pathways define key questions, red flags, and escalation triggers
 - Help structure the reasoning process
- Guideline rules
 - Important guideline logic encoded as rules (e.g., Datalog)
 - Used for red flags, contraindications, and escalation decisions
 - Ensures safety-critical checks are enforced
- Knowledge graph + Logic
 - Knowledge graph: medical concepts and relationships
 - Logic rules: check pathway conditions and safety constraints

CLINICAL KNOWLEDGE INTEGRATION

- Verification substrate
 - evaluate whether patient state satisfies guideline conditions
 - test whether reasoning steps remain consistent with knowledge
- Knowledge Graph (KG) + Logic enables controlled reasoning
 - KG stores concepts, relations, pathway nodes, and patient facts
 - rules evaluate pathway conditions, red flags, contraindications, and action permissions
 - neural reasoning proposes hypotheses using graph-retrieved context

THE SCIENCE OF SAFE (CLINICAL) AI

- From prediction to controlled reasoning
 - Early AI systems in medicine focused on prediction or question answering.
 - Diagnostic support requires something different: structured reasoning under uncertainty, where hypotheses evolve, evidence accumulates, and decisions are constrained by safety.
- The **neural–symbolic boundary** as a design variable
 - A key scientific question is where the boundary between neural inference and symbolic control should lie.
 - Placing this **boundary at action authorization** allows flexible reasoning while preserving deterministic control over safety-critical outcomes.
- Verification as a first-class component of inference
 - The emerging shift is from evaluating answers to evaluating reasoning processes.
 - Trustworthy systems require:
 - explicit intermediate reasoning artifacts
 - verification against structured evidence and clinical knowledge
 - rule-based enforcement of safety constraints
 - In this view, reasoning becomes an inspectable computational object, not an opaque model behavior.

THE SCIENCE OF SAFE (CLINICAL) AI

- Selective formalization of clinical knowledge
 - Medicine is too complex to encode entirely in rules, yet too safety-critical to leave entirely to probabilistic inference.
 - The scientific challenge is identifying which parts of reasoning must be deterministic:
 - red-flag detection
 - escalation conditions
 - contraindications
 - action permissions
 - Everything else, interpretation, hypothesis generation, exploratory reasoning, remains neural.
- Essentially
 - Differential diagnosis is a multi-regime reasoning problem.
 - Safe AI systems therefore require architectures that combine:
 - neural flexibility
 - symbolic verification
 - structured knowledge
 - explicit safety constraints
- The frontier of medical AI is not simply larger models, but architectures that make reasoning itself controllable, inspectable, and safe.

PRINCIPLES

- Decision agents must operate safely under **uncertainty**
- Neural–Symbolic **boundary at action authorization**
- **Formalize danger** rather than expertise
- Neural–Symbolic **disagreement** is inevitable yet desirable